Next-Day Stock Status Prediction in Fresh Retail

Anurag Thakur
2023IMG-008 Samyak Choudhary
2023IMG-044 Vibhor Kumar
2023IMG-054

Department of Information Technology
Bachelor of Technology Project (BTP)

August 2026

Problem Statement & Objective Metrics

Operational Challenge in Fresh Retail

- Perishable fresh items suffer from **rapid intra-day demand shocks** and supply replenishment lags.
- Unpredicted stockouts cause direct financial revenue loss, stock spoilage, and customer churn.

Predictive Target Formulation

- Forecast a **24-dim binary hourly status vector**: $\mathbf{Y} = [y_1, y_2, \ldots, y_{24}]^T \in \{0, 1\}^{24}$
- Covers active operational store hours (**6 AM to 10 PM**).
- $y_h = 1$ indicates SKU stockout during hour h.

Key Objective Performance Metrics

- **Hour-Level F1-Score** (Primary Metric): Evaluates precision vs. recall on positive stockout hours: $F1 = \frac{2 \cdot P \cdot R}{P+R}$
- **Stockout Recall**: Measures percentage of true operational stockout hours correctly detected ($R = \frac{TP}{TP+FN}$).

Literature Review & Related Work

Taxonomy of Time-Series Forecasting Literature

- **Recurrent Sequence Memory** (Hochreiter & Schmidhuber, 1997): *LSTM* maintains persistent cell state memory C_t for temporal sequence continuity.
- **Deep Time-Series Transformers** (Lim et al., 2021 [TFT]; Zeng et al., 2023 [PatchTST]): Self-attention mechanisms suffer from **overparameterization ($>$ 280k params)** on short sequences.
- **Linear Series Decomposition** (Zeng et al., 2023 [DLinear]; Challu et al., 2023 [N-HiTS]): Moving-average trend decomposition provides parameter-efficient baselines ($<$ 15k params).
- **Explainable AI in Retail** (Sundararajan et al., 2017 [IG]; Lundberg & Lee, 2017 [SHAP]): Attribution methods reveal feature-level importance to guide neural architecture design.
- **FreshRetail Benchmark Studies** (Same Dataset): Recent domain-specific works like *"A Coupled Hierarchical Architecture for Multi-Granularity Demand Forecasting"* and *"When Less Data Is Better: Eliminating Stockout-Induced Bias in Choice Estimation"* use this dataset to model complex stockout biases.

FreshRetailNet-50K Dataset Overview

Dataset Scale & Scope

- **Volume**: 50,000 store-product pairs across 898 stores.
- **Timeframe**: 90-day series of highly granular hourly data.
- **Products**: 865 diverse perishable SKUs.

Rich Feature Set

- **Sales & Stock**: Hourly sale amounts & exact stock status.
- **Events**: Discount rates, holiday flags, and activities.
- **Weather**: Precipitation, temp, humidity, and wind.

Why We Selected This Dataset

- **Pioneering Benchmark**: It is the first large-scale public dataset specifically designed for **censored demand estimation** (where out-of-stock items hide true customer demand) in the fresh retail domain.
- **Perishable Goods Focus**: Perfectly captures the unique, rapid intraday volatility of short-shelf-life items, containing ~20% organically occurring stockout data.

Data Selection & Splitting Strategy

Unit of Analysis & Filtering

- **Target Grain**: Predicted at the level of city_id + store_id + product_id.
- **Filtering Logic**: Fresh retail datasets contain thousands of dormant items. We select the **Top 15 SKUs** to focus on high-impact, stockout-prone goods.

SKU Statistical Ranking Formula

$$\text{Score} = 0.40 \times \text{Sales} + 0.25 \times \text{Stockout Hours}$$
$$+ 0.20 \times \text{Stockout Days} + 0.15 \times \text{Sales Std}$$

Train/Validation Split

- **Chronological Split**: Strict temporal separation to prevent data leakage.
- **Training Set**: Historical data up to $T_{\text{max}} - 15$ days (~ 2.5 months).
- **Validation Set**: Final 15 days of data.
- **Standardization**: `StandardScaler` fitted strictly on the training set to prevent lookahead bias.

Hourly vs. Daily Forecasting Paradigm

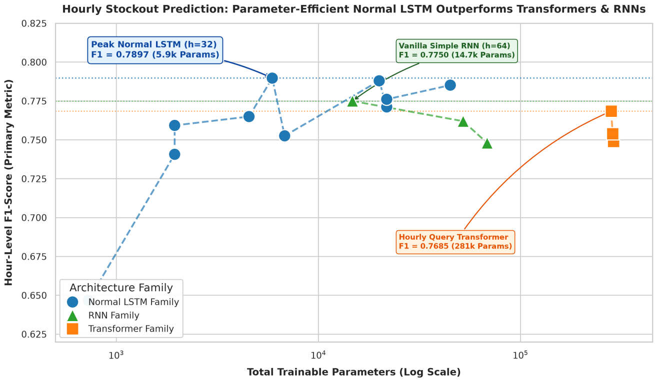
Traditional Daily Aggregation Paradigm

- **Input**: $L = 14$ days of historical daily metrics.
- **Output**: 1 scalar value (Total expected daily sales or binary stockout flag).
- **Loss of Information**: Store managers cannot determine *when* replenishment must occur.

Our 24-Hour Hourly Binary Paradigm

- **Input**: $L = 14$ days of SKU-store daily feature vectors.
- **Output**: 24 binary probabilities $\hat{\mathbf{Y}} \in [0, 1]^{24}$.
- **Actionable Insight**: Identifies peak risk hours (e.g. 11 AM lunch rush & 6 PM evening peak) for proactive restocking.

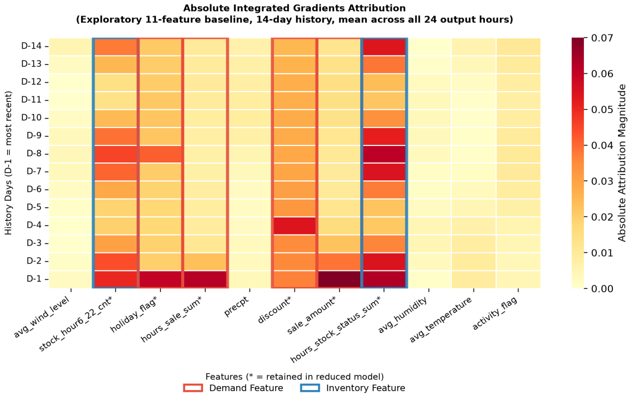
Model Family Comparison: Simple LSTMs Beat Heavy Transformers



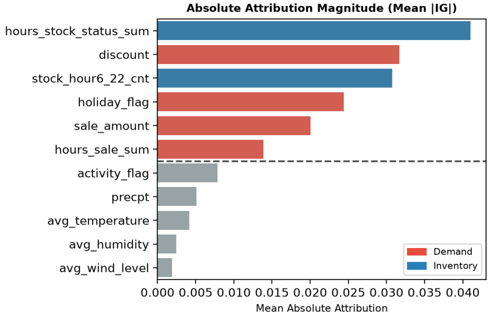
Key Empirical Findings

- **Parameter Efficiency**: Normal LSTMs (5.9k–44.9k params) reach **0.7852–0.7897 F1-Score**, matching or beating heavy Transformers.
- **Transformer Overparameterization**: PatchTST (285k params) & TFT (288k params) peak at **0.7540–0.7685 F1-Score**.
- **Why LSTMs Win**: Sequential state memory h_t naturally aligns with continuous hourly depletion better than permutation-invariant self-attention.

ML Explainability: Integrated Gradients Analysis



Attribution heatmap: all 11 features & 14-day history (D-1 = most recent)

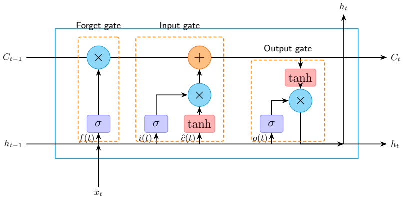


Feature importance: top 6 (highlighted) far outrank remaining 5

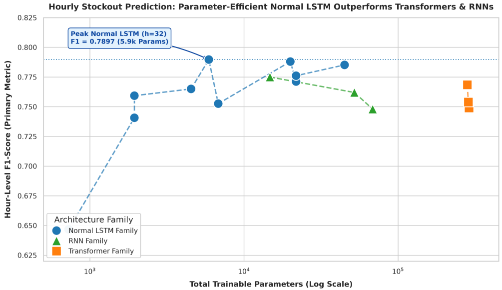
Observation → Selection

Absolute IG attribution for the exploratory 11-feature baseline (14-day history). The model relies most strongly on recent inventory and sales-related signals. This distinct temporal behavior motivates the Dual-Stream LSTM, which processes both groups independently before a fusion layer. Attribution indicates feature influence, not direct interaction or causality.

Limitations of Baseline LSTM



Standard LSTM Cell Architecture

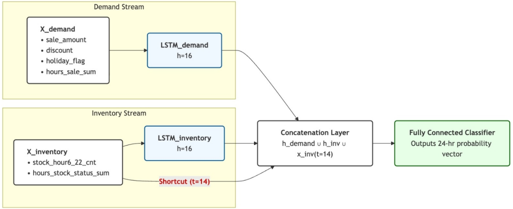


F1-Score across LSTM hidden size variants (h=8 to h=96)

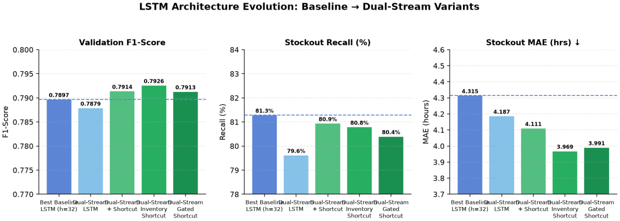
Why Scaling Alone Doesn't Help

- **F1 plateau** at ∼0.789 regardless of hidden size — more params yield no gain.
- Demand & inventory share hidden state h_t $\Rightarrow$ **feature cross-talk**.
- IG analysis (prev. slide) proves these signals need **dedicated streams**.

Our Fix: Dual-Stream Inventory Shortcut LSTM



Dual-Stream: Demand LSTM + Inventory LSTM + Day-14 Shortcut



F1, Recall & MAE: all dual-stream variants vs. baseline LSTM

Key Results

- **Inventory Shortcut**: F1 **0.7926**, MAE **3.97 hrs** at just 4,600 params.
- Decoupled streams consistently beat all baseline LSTM sizes.

Exploring Linear Decomposition: DLinear Models

Why Explore Linear Decomposition?

DLinear (Zeng et al., 2023) demonstrated that single-layer linear models decomposing time series into Trend and Seasonal components can outperform complex deep networks.

DLinear Series Decomposition

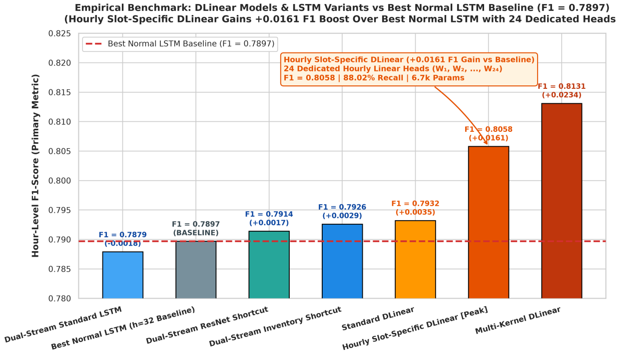
$$X_{\text{trend}} = \text{AvgPool}(X), \quad X_{\text{seasonal}} = X - X_{\text{trend}}$$

$$\mathbf{Y}_{\text{pred}} = W_{\text{trend}} X_{\text{trend}} + W_{\text{seasonal}} X_{\text{seasonal}}$$

Standard DLinear Performance

- **Parameters**: 6,768 parameters.
- **Validation F1-Score**: **0.7932**
    - Outperforms Normal LSTMs (**0.7897**)
    - Outperforms Enhanced LSTMs (**0.7926**)
    - Outperforms Transformers (**0.7540**)
- **Stockout Recall**: **87.44%**.

Architectural Innovation 2: Linear Changes in DLinear



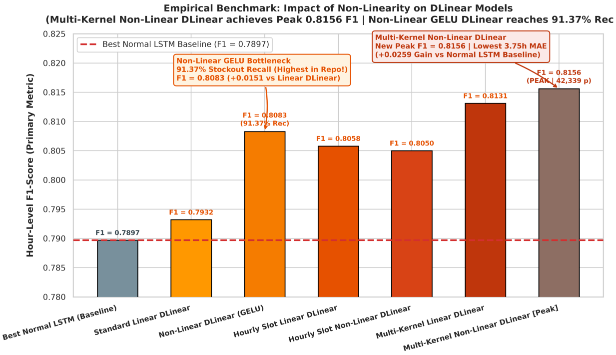
Hourly Slot-Specific Linear DLinear

- Uses **24 dedicated linear heads** $(W_1, \ldots, W_{24})$ for each operational hour.
- Resolves inter-hour gradient conflict between morning and evening rush hours.
- Boosts F1 to **0.8058** (+0.0161 over LSTM).

Multi-Kernel Linear DLinear

- Multi-scale moving average trend pooling ($k \in \{3, 5, 7\}$).
- Reaches **0.8131 F1-Score** with pure linear decomposition (13.5k params).

Architectural Innovation 3: Introducing Non-Linearity into DLinear



Non-Linear Activation Bottleneck

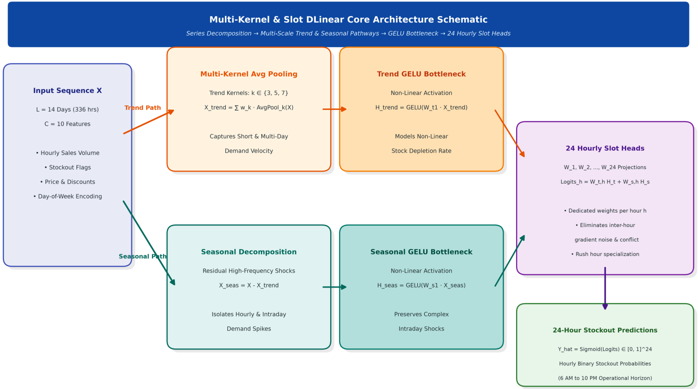
Adds GELU activation bottlenecks between decomposition and output projections:

$$\hat{Y} = W_{\text{trend},2}\text{GELU}(W_{\text{trend},1}X_{\text{trend}}) + \text{Seasonal}$$

Breakthrough Results

- **Non-Linear GELU DLinear**: Reaches **91.37% Stockout Recall** (Highest in Repo!) and **0.8083 F1**.
- **Multi-Kernel Non-Linear DLinear**: Reaches **0.8156 F1-Score** (New Peak F1 & lowest 3.75h MAE).

Architectural Schematic: Multi-Kernel & Slot DLinear



Core Mechanism

- **Trend Stream**: Multi-kernel moving average pooling ($k \in \{3, 5, 7\}$) + GELU bottleneck.
- **Seasonal Stream**: High-frequency residual stockout shocks ($X - X_{\text{trend}}$).
- **24 Slot Heads**: $W_1, \ldots, W_{24}$ dedicated projections resolve inter-hour gradient conflicts.

Master Empirical Validation Leaderboard

Rank	Architecture Name	Category	Params	F1-Score	Recall	MAE (h
1	**Multi-Kernel Non-Linear DLinear**	Non-Linear DLinear	42,339	**0.8156**	87.13%	**3.75**
2	**Multi-Kernel Linear DLinear**	Linear (Multi-Scale)	13,539	**0.8131**	85.22%	3.86
3	**Non-Linear GELU DLinear**	Non-Linear DLinear	21,168	**0.8083**	**91.37%**	4.20
4	Hourly Slot-Specific DLinear	Linear (24 Heads)	6,744	**0.8058**	88.02%	4.23
5	Standard Linear DLinear	Linear (Decomp)	6,768	**0.7932**	87.44%	4.28
6	Dual-Stream Inventory Shortcut	LSTM Family	4,600	**0.7926**	80.79%	3.96
Base	**Best Normal LSTM (h=32)**	Normal LSTM	5,912	**0.7897**	81.29%	4.31
7	Feature-Reduced Standard LSTM	Normal LSTM	19,992	0.7880	81.60%	4.58
8	Baseline Standard LSTM (h=96)	Normal LSTM	44,952	0.7852	80.40%	4.67
9	Vanilla Simple RNN (h=64)	RNN Family	14,744	0.7750	81.02%	4.23
10	PatchTST Patch Linear	Transformer	285,624	0.7540	81.90%	4.16

Conclusion & Operational Deployment Takeaways

Key Architectural Takeaways

1. **Non-Linearity Boosts DLinear**: GELU bottlenecks boost DLinear F1 to **0.8083** and Recall to a record **91.37%**.
2. **New All-Time Peak F1**: Multi-Kernel Non-Linear DLinear sets the highest F1-Score of **0.8156** and lowest MAE of **3.75 hours**.
3. **Feature Decoupling & Slot Heads**: Resolves inter-hour gradient noise and temporal cross-talk.

Deployment Recommendations

- **Primary Production Deployment**: **Multi-Kernel Non-Linear DLinear** (0.8156 F1, 3.75h MAE, 42.3k params).
- **Maximum Risk Mitigation**: **Non-Linear GELU DLinear** (91.37% Stockout Recall).
- **Future Scope**: Incorporate real-time intraday point-of-sale stream updates for dynamic threshold adjustment.